

PBMC3k Classification Model Comparison

Models were trained on 70% of cells, compared using the 20% validation partition, and evaluated on an untouched 10% test partition. Ranking is determined by validation macro F1, so all cell types contribute equally to model selection.

Table 1. Headline validation and test performance

Precision, recall, and weighted F1 weight each cell type by its number of cells. Macro F1 weights every cell type equally. AUC is multiclass one-vs-rest.

Rank	Model	Validation accuracy	Validation precision	Validation recall	Validation weighted F1	Validation macro F1	Validation weighted AUC	Test accuracy	Test precision	Test recall	Test weighted F1	Test macro F1	Test weighted AUC
1	XGBoost	0.928	0.931	0.928	0.927	0.928	0.993	0.902	0.901	0.902	0.900	0.893	0.991
2	SGD (gradient descent)	0.932	0.934	0.932	0.930	0.923	0.993	0.932	0.935	0.932	0.930	0.928	0.992
3	Logistic regression	0.926	0.928	0.926	0.924	0.915	0.995	0.932	0.935	0.932	0.931	0.934	0.994
4	Support vector machine	0.913	0.915	0.913	0.910	0.901	0.988	0.909	0.910	0.909	0.906	0.907	0.986
5	Extra Trees	0.907	0.913	0.907	0.902	0.892	0.992	0.905	0.913	0.905	0.901	0.892	0.993
6	Multinomial Naive Bayes	0.919	0.924	0.919	0.910	0.891	0.993	0.913	0.919	0.913	0.906	0.895	0.992
7	MLP	0.902	0.903	0.902	0.896	0.882	0.991	0.902	0.902	0.902	0.897	0.885	0.989
8	Random forest	0.911	0.918	0.911	0.898	0.869	0.993	0.890	0.903	0.890	0.875	0.848	0.991
9	K-nearest neighbors	0.455	0.624	0.455	0.399	0.391	0.782	0.443	0.581	0.443	0.367	0.302	0.766

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Table 2. Diagnostic metrics and tuning details

Balanced accuracy and macro metrics emphasize performance across classes. Training CV values summarize three-fold cross-validation conducted only inside the training partition. Best parameters are the settings selected for each model family.

Rank	Model	Validation balanced accuracy	Validation macro precision	Validation macro recall	Validation macro AUC	Test balanced accuracy	Test macro precision	Test macro recall	Test macro AUC	Training CV macro F1 mean	Training CV macro F1 SD	Fit time (seconds)	Best parameters
1	XGBoost	0.915	0.952	0.915	0.995	0.890	0.898	0.890	0.991	0.890	0.017	74.3	{"classifier__learning_rate": 0.1, "classifier__max_depth": 6}
2	SGD (gradient descent)	0.909	0.949	0.909	0.994	0.916	0.950	0.916	0.991	0.882	0.019	7.9	{"classifier__alpha": 0.0001, "classifier__loss": "log_loss"}
3	Logistic regression	0.899	0.944	0.899	0.996	0.924	0.950	0.924	0.993	0.897	0.007	2.9	{"classifier__C": 0.1}
4	Support vector machine	0.884	0.936	0.884	0.990	0.899	0.928	0.899	0.987	0.866	0.007	0.9	{"classifier__C": 0.01}
5	Extra Trees	0.873	0.943	0.873	0.994	0.866	0.948	0.866	0.994	0.885	0.009	17.8	{"classifier__max_features": "sqrt", "classifier__min_samples_leaf": 2}
6	Multinomial Naive Bayes	0.881	0.949	0.881	0.993	0.888	0.944	0.888	0.991	0.863	0.004	0.2	{"classifier__alpha": 1.0}
7	MLP	0.869	0.919	0.869	0.991	0.875	0.916	0.875	0.988	0.836	0.004	3.2	{"classifier__alpha": 0.0001, "classifier__hidden_layer_sizes": [128]}
8	Random forest	0.863	0.947	0.863	0.995	0.846	0.941	0.846	0.991	0.826	0.002	8.7	{"classifier__max_depth": 20, "classifier__min_samples_leaf": 2}
9	K-nearest neighbors	0.384	0.686	0.384	0.757	0.313	0.542	0.313	0.781	0.303	0.013	0.4	{"classifier__n_neighbors": 5}

Selected model: XGBoost (highest validation macro F1 = 0.928). Test results were not used to choose the model.